

Lithium Carbonate Demand (1)

Energy Storage Growth

碳酸锂需求展望（一）：储能增速

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Abstract 摘要

Industrial Policies Support Rapid Expansion of Global Energy Storage. The implementation of capacity pricing mechanisms in China and expanding opportunities for spot market arbitrage have significantly improved returns on energy storage projects. Meanwhile, the extension of ITC subsidies in the US, coupled with favorable policy environments and robust project pipelines in Europe and emerging markets, has driven global demand **beyond expectations**. Under pessimistic, neutral, and optimistic scenarios, global energy storage installations are projected to reach **419, 449, and 499 GWh** in 2026, representing **YoY growth of 41%, 52%, and 68%, respectively**.

Risk factors: 1) Policy implementation effect is lower than expected; 2) External trade restrictions are escalated; 3) Sales volume is lower than expected; 4) Exports are lower than expected.

政策向好支撑全球储能高速增长。 中国容量电价机制利好与现货套利机会提高储能项目收益率，美国 ITC 补贴延续、欧洲及新兴市场政策利好叠加项目储备较多，全球储能需求或呈现超预期增长。

悲观、中性、乐观三种情景下，预测 2026 年全球储能装机将达 419/449/499GWh，同比增长 41%/52%/68%。

风险提示：1) 政策落地效果不及预期； 2) 外部贸易限制升级； 3) 销量不及预期； 4) 出口不及预期。

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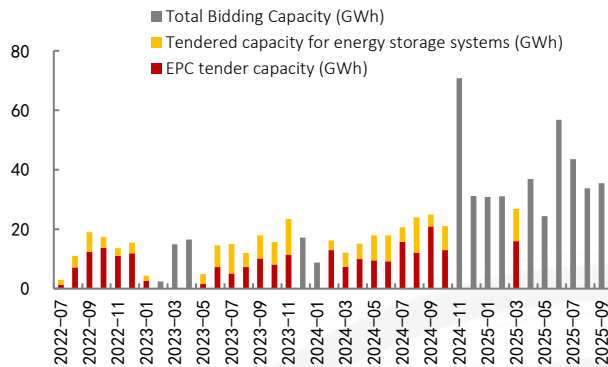
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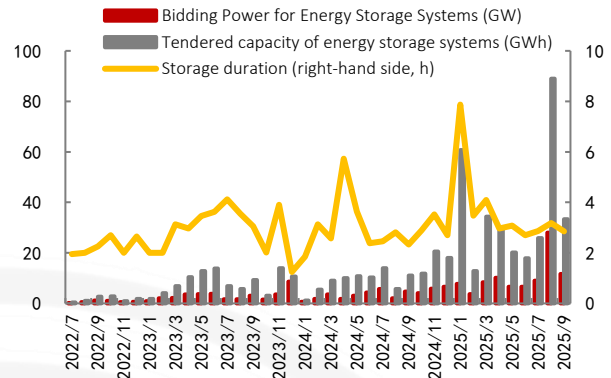
In 2025, both domestic and international energy storage demand has shown growth exceeding expectations. In China, energy storage projects have experienced robust growth since the third quarter of the year, driven by the advancement of power market reforms that have clarified the capacity pricing mechanism for energy storage. The increasing proportion of renewable energy generation has also widened the peak-to-valley price differential, making the profit model for energy storage increasingly clear. The high return on investment has attracted rapid growth in energy storage projects, with large utilities & EP construction groups showing strong demand for energy storage procurement, especially on the grid side, where demand has surged dramatically. In the US, The Inflation Reduction Act (IRA) has extended the ITC subsidy and adjusted supply chain review requirements to alleviate the current tight supply of energy storage cells. These policy adjustments have sustained the enthusiasm for energy storage investments and continued sequential growth in installations; clean energy goals in some states have further boosted the demand for photovoltaic systems paired with energy storage. In Europe and emerging markets, tailwinds from favorable policies and a substantial pipeline of projects have contributed to global energy storage demand growing beyond expectations.

1. China: Improved Returns on Energy Storage Projects, Installation Prospects Look Promising

In the third quarter of 2025, a dense release of reserve projects maintained high growth in tendering scale. In the northwest provinces with active utilities & EP markets, the capacity tariff mechanism significantly improved the economic viability of energy storage, leading to strong demand from large power construction groups. According to incomplete statistics, the total tendering scale for energy storage in the first three quarters of 2025 reached 87.65GW/319.55GWh, representing a YoY increase of 37%/103%; the main drivers were the sequential release of more than 30 GWh-level energy storage tendering projects in Xinjiang, Inner Mongolia, and Gansu provinces. From the bidding data, according to incomplete statistics, **the scale of successful bids for China's energy storage systems in the third quarter of 2025 was as high as 48.7GW/148.1GWh, up 292%/382% YoY, with robust demand in Xinjiang, Inner Mongolia, and Ningxia.** Notably, the scale of successful bids in Aug saw a significant increase, primarily due to the completion of tendering for 12 large individual GWh-level projects and the implementation of the 25GWh energy storage system procurement order by China Energy Engineering Corporation.

Chart 1: China's ESS Tender Scale

Sources: Xun Entropy, CITIC Futures

Chart 2: China's ESS Bidding Scale

Sources: Xun Entropy, CITIC Futures

The capacity leasing and spot market arbitrage model maintains high returns, supporting the mid-to-long-term rapid development of energy storage. Capacity leasing and spot market arbitrage have improved the return on investment for energy storage projects, attracting investor interest. In regions where the current peak-to-valley price difference remains at Rmb150-200/MWh, the return on energy storage projects can still reach 6.2%-9.8%, making them highly attractive. For areas with a peak-to-valley price difference of Rmb100/MWh, the current returns on energy storage are also in the range of 4.5%-6.8%. With the addition of capacity tariffs and other incentive policies in the future, these regions also show strong potential for development. As the proportion of new energy power entering the market gradually increases, **the peak-to-valley price difference is expected to widen, bringing new opportunities for energy storage participation in spot market arbitrage.**

The gradual improvement of China's energy storage capacity tariff mechanism is expected to further enhance the profitability of energy storage. The energy storage capacity tariff mechanism is a quasi-market mechanism that provides fixed compensation based on the effective capacity of energy storage, with the core aim of ensuring the recovery of fixed costs. Taking the current minimum capacity tariff of RMB 100/kW·year as an example, under the condition that the EPC price of energy storage remains at Rmb1.05 per Wh, the annual capacity tariff revenue can reach 2.4%-4.8% of the cost; in regions with high capacity tariffs such as Inner Mongolia and Gansu, the annual capacity tariff revenue can exceed 10% of the cost, significantly improving the return on investment for energy storage. In the future, China plans to pilot capacity markets in regions where conditions permit, promoting the realization of energy storage capacity value through market mechanisms.

Table 3: Average Returns from Participating in Power Spot Market Arbitrage in Some Provinces and Cities (CNY/MWh, per transaction)

Independent Storage Return on Investment		Average peak-to-valley price difference (unit: Rmb/kWh*cycle)						
		0.1	0.15	0.2	0.25	0.3	0.35	0.4
Average rental income (Unit: Rmb/kWh · year)	50	1.90%	3.80%	5.60%	7.20%	8.70%	10.20%	11.60%
	100	4.50%	6.20%	7.80%	9.30%	10.70%	12.10%	13.50%
	150	6.80%	8.30%	9.80%	11.30%	12.60%	14.00%	15.30%
	200	8.90%	10.40%	11.80%	13.10%	14.50%	15.80%	17.00%
	250	10.90%	12.30%	13.60%	14.90%	16.20%	17.50%	18.70%
	300	12.80%	14.10%	15.40%	16.70%	17.90%	19.20%	20.40%
	350	14.60%	15.90%	17.10%	18.40%	19.60%	20.80%	22.00%

Sources: Lambda, CITIC Futures

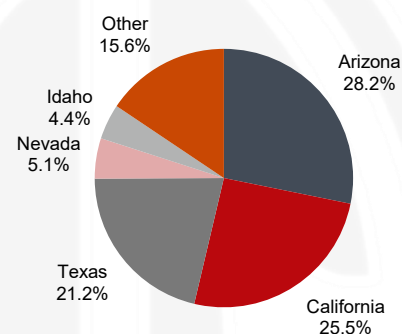
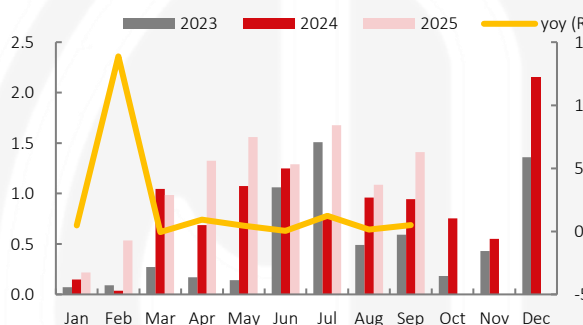
2. US: Subsidy Policies Continue, Energy Storage Investments Return

The Inflation Reduction Act (IRA) retains subsidies and adjusts supply chain review requirements, expecting that the US energy storage cell supply chain issues will ease during 2026-27. In the One Big Beautiful Bill Act, the ITC for large-scale storage begins to phase down from 2034, with rates of 75%, 50%, and 0% in 2034, 2035, and 2036, respectively. The stable ITC subsidy expectations ensure long-term investment returns in energy storage, encouraging reinvestment. Regarding supply chain reviews, the cost share of sensitive entities must not exceed 60%, 55%, 50%, and 45% in 2026, 2027, 2028, and 2029, respectively, and it must drop to 40% after 2030 to qualify for subsidies. Currently, US energy storage still relies heavily on Chinese-produced cells, with cell costs accounting for about 50% of total costs, providing a window for US energy storage companies to achieve a smooth transition in their supply chains. In 3Q25, the US added 4.17 GW of new large-scale storage capacity, representing a 57% increase YoY; in the first three quarters of 2025, the US added 10.08 GW of new large-scale storage capacity, marking a 46% increase YoY.

Clean energy goals in some states continue to drive demand for photovoltaic (PV) paired storage. Taking Arizona as an example, the state's storage demand primarily stems from the need to pair storage with solar power installations. Arizona has already built over 8GW of solar power capacity, which accounts for 20% of the state's utilities & EP mix, ranking fifth in the nation. The high proportion of solar power has spurred the demand for paired storage. Additionally, Arizona Public Service Company (APS), the largest utility company in the state, has proposed achieving 100% clean energy supply by 2050, with an interim goal of 65% clean energy by 2030. To achieve these targets, significant storage capacity will be required to stabilize the fluctuations in renewable energy supply.

Against the backdrop of declining yields and a slowdown in wind and solar installations, the medium-to-long-term leading indicators for the U.S. energy storage sector have begun to stagnate. According to EIA data from September, the capacity of energy storage projects under construction was recorded at 23.15 GW, representing a 31% year-over-year increase but only a 5% marginal gain month-over-month, while "planned but not yet under construction" projects reached 43 GW—an 81% year-over-year jump that remains largely flat compared to the start of the year. Neither metric has shown significant sequential growth entering 2025, a trend driven on one hand by rising tariffs and compressed yields, and on the other by the anticipated impact of Trump's policies restricting renewable energy development, which has dampened long-term demand for integrated storage solutions.

Chart 4: US Monthly Utility ESS Installation Scale(GW) Chart 5: Utility-scale ESS share in US states (%)



Sources: EIA, CITIC Futures

Sources: EIA, CITIC Future

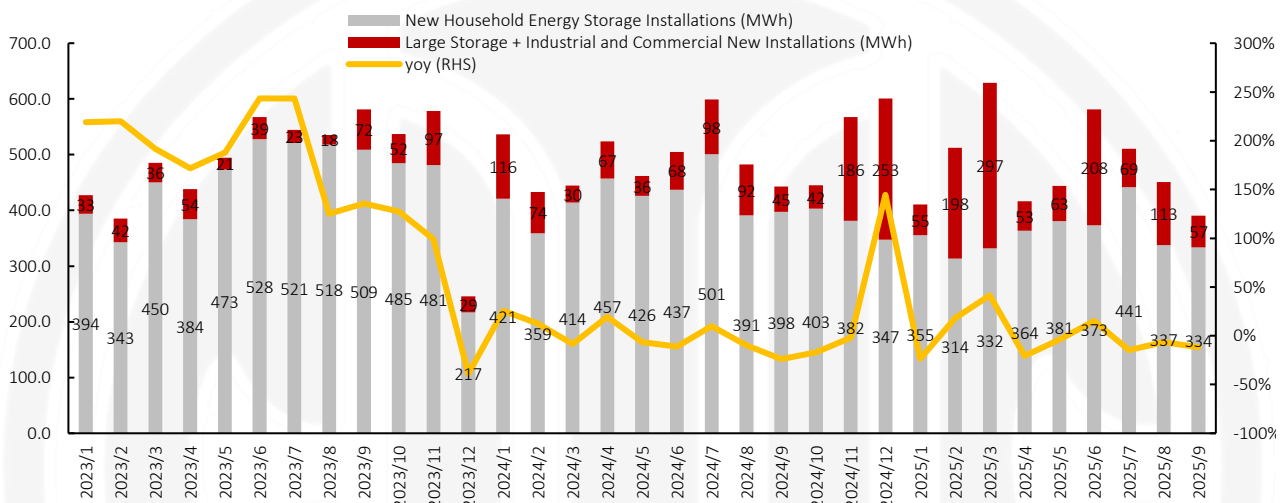
3. Europe and Emerging Markets: Large-scale storage in Europe continues to grow at a high rate, while emerging markets have become the engine of growth

Storing demand in Europe and emerging markets is showing a high growth trend, with policy support and market mechanism improvements being the core driving forces. In Europe, although residential storage demand decreased by 15% YoY, the large-scale storage market maintained strong growth. Germany's large-scale storage and commercial and industrial (C&I) storage saw a significant increase of 78% YoY due to policy adjustments. The UK has planned projects exceeding 68GWh and has a mature business model, while Italy's MACSE auction mechanism has activated the market. It is expected that Europe's installed storage capacity will reach 30GWh by 2026.

In 2025, energy storage demand in emerging markets such as Australia and the Middle East showed explosive growth. Strong overseas demand drove Chinese energy storage orders abroad, and by the end of the third quarter of 2025, Chinese companies had signed project scales

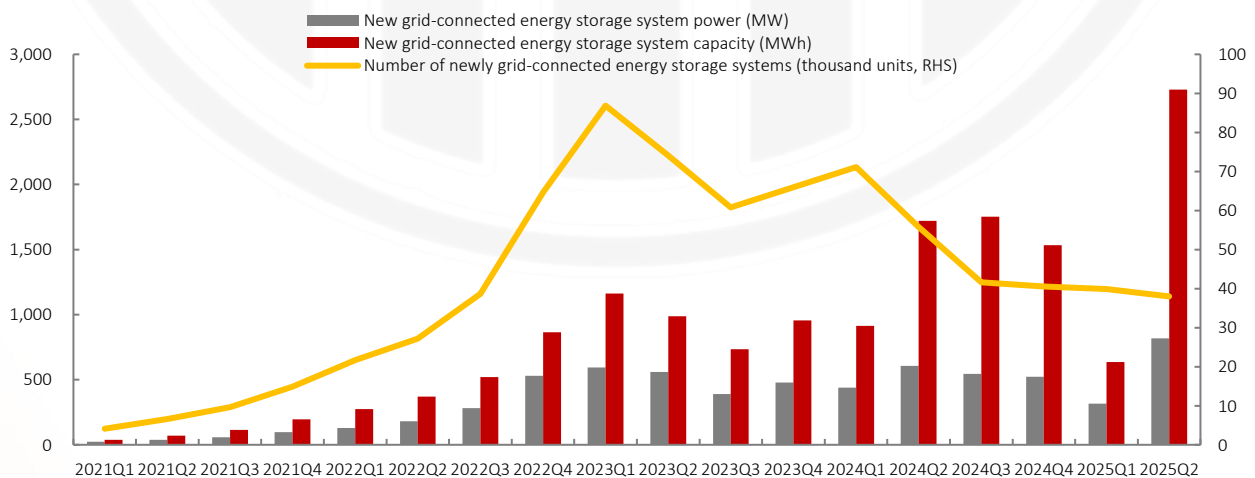
totaling 48.08 GWh with **Europe** (an increase of 15.77 GWh in 3Q25), 43.21 GWh with **Australia** (an increase of 10.9 GWh in 3Q25), and 40.06 GWh with the **Middle East** (an increase of 2.51 GWh in 3Q25). Other significant markets included **India** (with a cumulative total of 13.25 GWh in the first three quarters) and **Chile** (with a cumulative total of 10.81 GWh in the first three quarters). However, the construction cycle for energy storage projects in emerging markets is typically 2-3 years, leading to slower installation speeds. Overall, **emerging markets are expected to become a key driver of global energy storage growth in the future, continuously boosting global energy storage.**

Chart 6: Germany's monthly added energy storage capacity (MWh)



Sources: ISRA, CITIC Futures

Chart 7: Italy's Single Quarter Energy Storage Installation Scale and Quantity



Sources: Lambda, CITIC Futures

4. Energy Storage Demand Forecast Under Three Scenarios

We believe that China and emerging markets will have strong performance in 2026, while the European and American markets will maintain stable growth.

In 2026, China's newly installed capacity is expected to reach 230GWh. In 2026, China's energy storage installation volume is projected to maintain a growth rate similar to recent tender scales. Given that the **tender project capacity increased by 65% in the first three quarters of 2025**, and with China's energy storage installations totaling around 140GWh in 2025, the installation scale for 2026 is expected to reach 230GWh, growing by approximately 90GWh, which will **contribute more than half of the global installations**.

In 2026, the US market installed capacity is expected to be approximately 52GWh. The US EIA predicts that the large-scale storage installed capacity for 2026 will be 19.5GW/58.5GWh. Assuming the completion rate of energy storage projects in the US increases to 85% in 2026, the large-scale storage installed capacity will approach 50GWh. Commercial and industrial storage, along with residential storage, are expected to maintain demand levels similar to those in 2025 (2GWh). Therefore, the total installed capacity will be 52GWh, **representing an 11% increase YoY**.

It is expected that the installation volume in Europe and emerging markets will be approximately 167GWh in 2026. The growth in the European market in 2026 will primarily come from the expansion of large storage markets in various countries, with Germany and Italy expected to contribute increments of 4GWh and 3GWh, respectively, while increments from other countries will be counted under the emerging markets. By the end of 2025, the new tender scale in emerging markets is estimated to be around 300GWh. Considering the relatively slow construction progress, about 40% of this can be realized in 2026, corresponding to an installation volume of 120GWh. Adding the cumulative pre-project installations of around 15GWh, including the Tabuk and Hail provinces energy storage projects in Saudi Arabia (4GWh), the third and fourth rounds of the AEMO system projects in Australia (4+3GWh), the Victoria and South Australia 600MW/2.4GWh project, and the Colbún project in Chile (228MW/912MWh), the total installation volume for Europe and emerging markets is roughly 167GWh, **representing a YoY increase of 52%**.

It is expected that global energy storage installations will increase to 449GWh in 2026, with energy storage cell production adjusted upward to 756GWh. This latest forecast is significantly higher than our market assumptions at the beginning of 2025 (previously predicted at 369GWh and 513GWh), **corresponding to an additional demand for lithium carbonate of 150kt**. This could have a significant impact on the supply and demand dynamics of lithium carbonate in 2026.

In 2025, supported by policies both domestically and internationally and driven by market demand, the development of the energy storage market has exceeded expectations. We have revised our predictions for 2026 energy storage installations upwards, primarily due to two factors: **first, the procurement in China's third quarter has consistently surpassed expectations**,

demonstrating the robust potential of the domestic energy storage market under capacity tariff incentives; **second, demand in overseas emerging markets (including some European countries) has entered a period of rapid growth**, with continuous bidding for large projects. Although traditional markets such as the US and Germany have shown relatively stable performance, under the impetus of these two new drivers, **the overall global demand for energy storage in 2026 will grow by more than 50%**, thus boosting the lithium battery-related materials into a high-prosperity cycle.

It is expected that the capacity deployment rhythm of energy storage projects will be relatively even in 2026. According to historical patterns, assuming the demand peak for 2026 advances to 2Q26 and the peak demand volume approaches 75-80 GWh, there is still a good chance that the operating rate of the cell industry will reach above 78%.

At the same time, we have noticed that there is some divergence in the market regarding the growth rate of energy storage, with the **main point of contention being the rapidly growing China domestic energy storage market**. To explore possible scenarios for energy storage demand both domestically and internationally in 2026, we have made two assumption-based projections based on different domestic energy storage tender growth rates.

a) Aggressive Scenario: Assuming the domestic energy storage bidding growth rate reaches 100% from 2H25 to 1H26, a growth level close to the current bidding growth rates in high-yield and high-capacity tariff regions such as Xinjiang and Inner Mongolia, then the new domestic energy storage installation capacity in 2026 will approach 280GWh, driving the global energy storage installation capacity to nearly 500GWh, and cell production to approximately 850GWh, corresponding to a lithium carbonate demand of 0.51mn t.

b) Conservative Scenario: Assuming the domestic energy storage bidding growth rate decreases to 40% in 2026, which is the growth rate level of 3Q25, then the additional installed capacity of domestic energy storage in 2026 would be approximately 200GWh, the global energy storage installed capacity would be adjusted to 419GWh, and the cell production would approach 700GWh, corresponding to a lithium carbonate demand of 0.42mn tons.

Chart 8: Global Energy Storage New Installation Forecast from 2021 to 2026

Project	Unit	2021	2022	2023	2024	2025E	2026E		
							Conservative	Moderate	Aggressive
Chinese Installed Energy Storage Capacity Additions	GWh	5	16	47	103	141	200	230	280
Overseas Installed Energy Storage Capacity Additions	GWh	25	50	70	104	157	219		
Includes: US	GWh	10	13	26	37	47	52		
Includes: Europe and Emerging Markets	GWh	15	37	44	67	110	167		
Global Installed Energy Storage Capacity Additions	GWh	30	67	117	207	297	419	449	499
yoy	%		122%	75%	77%	43%	41%	52%	68%
Global Energy Storage Cell Demand	GWh	66	145	193	347	499	700	756	850
yoy	%		119%	33%	80%	44%	40%	52%	70%
Cell inventory buffer factor		2.21	2.2	1.65	1.68	1.68	1.67	1.68	1.70
Global Lithium Carbonate Equivalent (LCE) Demand from the Energy Storage Sector	kt	40	87	116	208	299	420	454	510

Sources: SMM, CITIC Futures

The production landscape of energy storage cells is dominated by China, with an expected share of nearly 97% of global production in 2026, corresponding to a lithium carbonate demand of 440kt. The localization rate of energy storage cells in China increased from 64% in 2021 to 96%-97% in 2024, marking a complete shift from the "tripod of China, Japan, and South Korea" to a China-led sector. Overseas battery giants (such as LG and Samsung SDI) have effectively exited the mainstream competition in energy storage cell manufacturing due to their commitment to the ternary lithium route and high costs. In contrast, China has fully mastered the lithium iron phosphate (LFP) industry chain, from upstream lithium salt processing to cathode and anode materials, giving it absolute pricing power and cost control. Moreover, Chinese companies have taken the lead in advancing the transition from 280Ah to 314Ah and beyond 500Ah, with a specialization iteration speed for energy storage scenarios far surpassing that of overseas competitors, thereby establishing a technological generation gap. In the future, the global energy storage market will likely maintain a division of labor characterized by "Chinese-made cells + overseas integration/assembly." It is anticipated that China's share of global energy storage cell production will remain at a high level of nearly 97%; under a neutral scenario, China's energy storage cell production in 2026 is expected to correspond to a lithium carbonate demand of 440kt.

Chart 9: Battery Cell Production and Corresponding LC Demand from 2021 to 2026

Project	Unit	2021	2022	2023	2024	2025E	2026E		
							Conservative	Moderate	Aggressive
Global Energy Storage Cell Demand	GWh	66	145	193	347	499	700	756	850
Includes: China	GWh	42	130	185	335	482	676	733	821
Chinese Share	%	64%	90%	96%	97%	97%	97%		
Global Lithium Carbonate Equivalent (LCE) Demand from the Energy Storage Sector	kt	40	87	116	208	299	420	454	510
Includes: China	kt	25	78	111	201	289	405	440	492

Sources: SMM, CITIC Futures

We have further broken down the production rhythm of energy storage cells on a monthly basis for 2026 (see Chart 10) to comprehensively analyze the supply and demand dynamics of lithium carbonate in 2026, aiming to identify potential mismatches within the year.

Chart 10: Monthly Forecast for Global and China Energy Storage Cell Production in 2026 and Corresponding LC Demand Estimation

2026 (Neutral Scenario)	Energy storage cell production (GWh)		Corresponding lithium carbonate demand (kt)	
	Global	China	Global	China
Jan	51.0	49.5	30.6	29.7
Feb	53.0	51.4	31.8	30.8
Mar	55.0	53.4	33.0	32.0
Apr	58.0	56.3	34.8	33.8
May	61.0	59.2	36.6	35.5
Jun	64.0	62.1	38.4	37.2
Jul	67.0	65.0	40.2	39.0
Aug	70.0	67.9	42.0	40.7
Sep	70.0	67.9	42.0	40.7
Oct	69.0	66.9	41.4	40.2
Nov	69.0	66.9	41.4	40.2
Dec	69.0	66.9	41.4	40.2
Total for 2026	756	733.4	453.6	440.0

Sources: SMM, CITIC Futures

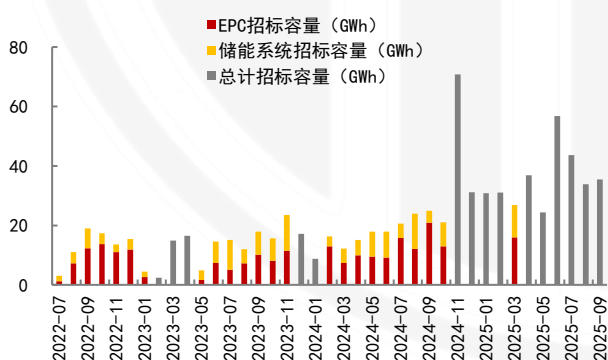
Appendix: Chinese Version

2025 年，国内外储能需求呈现超预期增长态势。在中国，储能项目在今年三季度以来增长旺盛，电力市场改革推进等明确了储能容量电价机制，新能源发电比例提高也扩大了峰谷价差，使得储能盈利模式逐渐明晰，较高的收益率吸引储能项目快速增长，大型电建集团储能采招需求旺盛，尤其电网侧储能呈现出井喷式高增。在美国，大而美法案延续了 ITC 补贴，并调整了供应链审查要求，以缓解当前储能电芯供应紧张情况，政策调整使得储能投资热度延续、装机继续环比增长；部分州的清洁发展目标持续拉动光伏配储需求。欧洲及新兴市场政策利好叠加项目储备较多，全球储能需求呈现超预期增长。

1. 中国：储能项目收益率提升，装机景气度可期

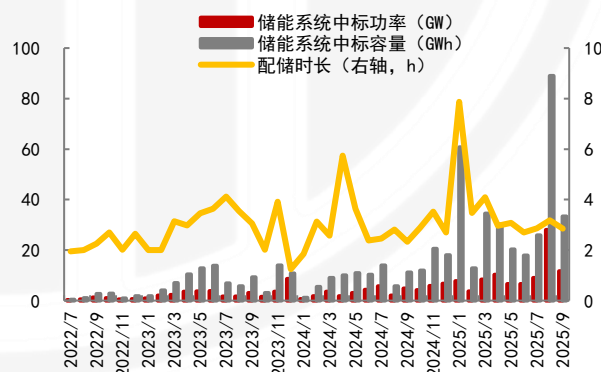
2025 年三季度储备项目密集释放，招标规模维持高增长。在电力市场活跃的西北省份，容量电价机制大幅提高储能经济性，大型电建集团储能采招需求旺盛。从招标数据看，据不完全统计，2025 年前三季度储能招标总规模达到 87.65GW/319.55GWh，同比增长 37%/103%；主要动力来自于新疆、内蒙、甘肃等省份多达 30 个 GWh 级储能招标项目陆续释放。从中标数据看，据不完全统计，2025 年三季度中国储能系统中标规模高达 48.7GW/148.1GWh，同比+292%/+382%，新疆、内蒙、宁夏等市场需求强劲。其中 8 月中标规模高增，主要受益于 12 个 GWh 级大型单体项目完成采招，以及中能建 25GWh 储能系统集采订单落地。

图表1：中国新型储能招标容量



资料来源：寻嫡 中信期货研究所

图表2：中国新型储能中标规模



资料来源：寻嫡 中信期货研究所

容量租赁与现货套利模式保持较高收益率，支持储能中长期高速发展。容量租赁与现货套利提高了储能项目收益率，吸引投资兴趣，对于当前峰谷价差仍在 150-200 元/MWh 的地区，储能项目的收益率仍可达 6.2%-9.8%，具备较强的吸引力；对于部分峰谷价差在 100 元/MWh 的地区，目前储能收

益也在 4.5%-6.8%，未来叠加容量电价等激励政策，也具备较强的发展潜力。未来随着新能源电力入市比例逐步提升，**峰谷价差有望走阔，为储能参与现货市场套利带来新的机遇。**

图表3：部分省市储能参与电力现货市场套利的年度平均收益（元 / MWh，单次）

独立储能收益率		平均峰谷价差（单位：元/kWh*次）						
		0.1	0.15	0.2	0.25	0.3	0.35	0.4
平均租赁收入 (单位：元/ 度·年)	50	1.90%	3.80%	5.60%	7.20%	8.70%	10.20%	11.60%
	100	4.50%	6.20%	7.80%	9.30%	10.70%	12.10%	13.50%
	150	6.80%	8.30%	9.80%	11.30%	12.60%	14.00%	15.30%
	200	8.90%	10.40%	11.80%	13.10%	14.50%	15.80%	17.00%
	250	10.90%	12.30%	13.60%	14.90%	16.20%	17.50%	18.70%
	300	12.80%	14.10%	15.40%	16.70%	17.90%	19.20%	20.40%
	350	14.60%	15.90%	17.10%	18.40%	19.60%	20.80%	22.00%

资料来源：Lambda 中信期货研究所

我国储能容量电价机制的逐步完善，也有望进一步增厚储能收益。储能容量电价机制是按储能有效容量给予固定补偿的准市场化机制，核心是保障固定成本回收。以当前容量电费最低标准 100 元/kWh·年为例，在储能 EPC 价格维持 1.05 元/Wh 的情形下，每年容量电价收入可以达到成本的 2.4%-4.8%；在内蒙、甘肃等高容量电价地区，每年容量电价收入可以占成本的 10%以上，显著提升储能收益率。未来我国将在具备条件地区试点容量市场，促进储能容量价值通过市场化方式兑现。

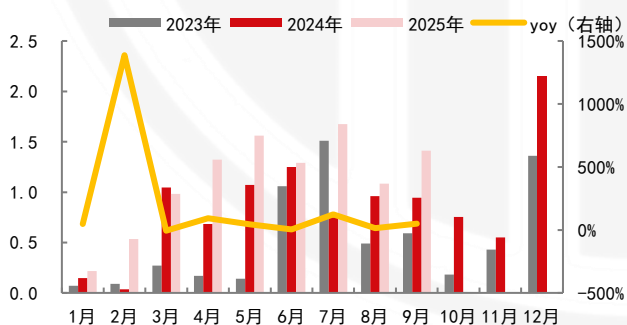
2. 美国：补贴政策延续，储能投资回归

大而美法案保留补贴，并调整供应链审查要求，预计 2026-27 年美国储能电芯供应链问题将有所缓解。在大而美法案中，大储 ITC 退坡自 2034 年开始，2034/35/36 年分别为原来的 75%/50%/0%，ITC 补贴预期稳定，使得储能的长期投资收益得到保障，投资回归。在供应链审查方面，2026-29 年敏感实体成本占比不超过 60%/55%/50%/45%，30 年后降至 40%即可获得补贴。当前美国储能仍依赖中国产电芯，且电芯成本占比约为 50%，这一政策为美国储能企业供应链的平稳过渡争取了时间窗口。2025 年三季度，美国大储新增装机 4.17GW，同比增长 57%；前三季度美国大储新增装机 10.08GW，同比增长 46%。

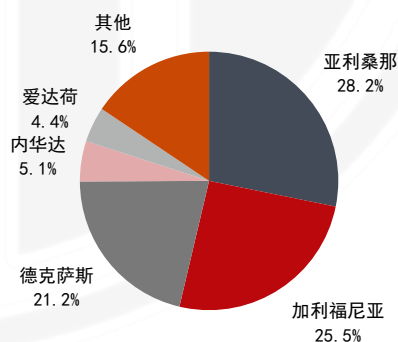
部分州的清洁发展目标持续拉动光伏配储需求。以亚利桑那州为例，该州的储能需求主要来自太阳能装机配储需求，已建成超过 8GW 太阳能装机容量，太阳能已占该州电力结构的 20%，排名全美第五，高比例的太阳能供电催生配储需求。同时，当地最大电力公司亚利桑那公共服务公司（APS）提出，到 2050 年实现 100%清洁能源供电，中期目标为 2030 年清洁能源占比 65%，因此需要大量配置储能以平抑新能源电力波动。

在收益率下滑叠加光伏风电装机放缓的背景下，美国储能项目景气度中长期指标出现滞涨。根据 EIA，9 月统计的建设中的储能项目规模录得 23.15GW，同比增长 31%，环比增长仅 5%；尚未开工但有意向开发的项目录得 43GW，同比增长 81%，较年初基本持平。两者在进入 2025 年后均未出现明显环比提升，一方面是由于在关税增长和收益率下滑，另一方面则由于特朗普限制光伏与风电发展，导致配储需求中长期增长放缓。

图表4：美国单月大型储能新增装机规模（GW）



图表5：美国各州公用事业级储能占比（%）



资料来源：EIA 中信期货研究所

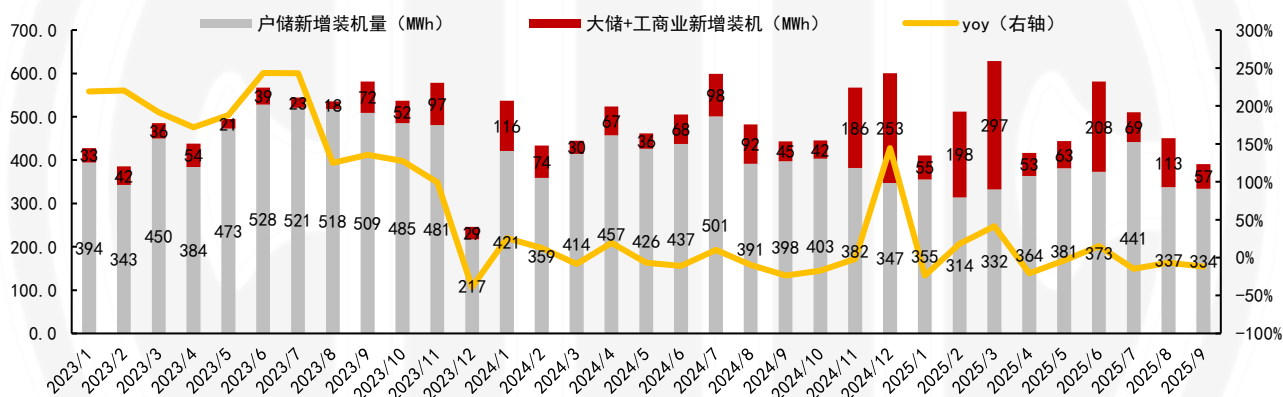
资料来源：EIA 中信期货研究所

3. 欧洲及新兴市场：欧洲大储保持高增，新兴市场成为增长引擎

欧洲及新兴市场储能需求呈高增态势，政策支持与市场机制完善为核心驱动力。欧洲方面，虽然户储需求同比下滑 15%，但大储市场保持强劲增长，德国大储及工商业储能因政策调整同比大增 78%，英国规划超 68GWh 项目且商业模式成熟，意大利 MACSE 拍卖机制激活市场。预计 2026 年欧洲储能装机将达 30GWh。

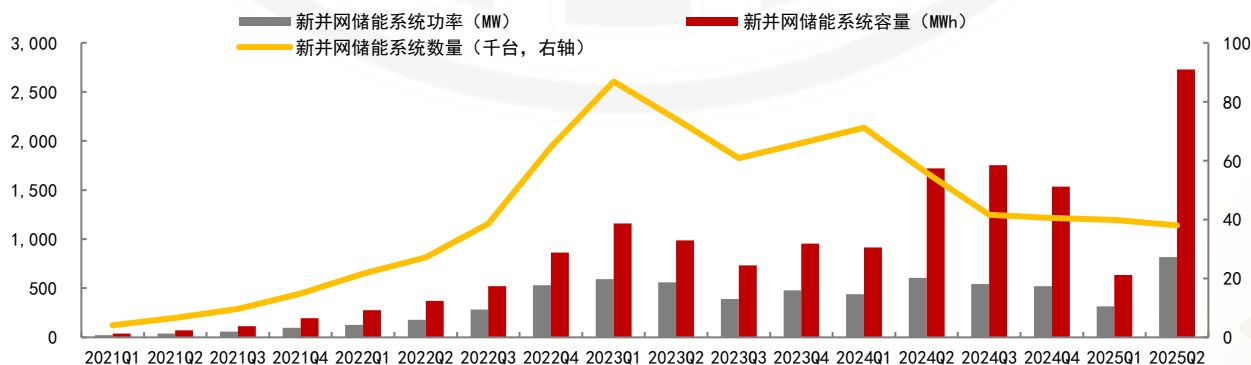
2025 年，澳洲、中东等新兴市场储能需求呈现爆发态势。强劲的海外需求拉动中国储能海外订单，截止 2025 年三季度，中国企业与欧洲签订项目规模 48.08GWh（Q3 增加 15.77GWh）、与澳大利亚签订项目规模 43.21GWh（Q3 增加 10.9GWh）、与中东签订项目规模 40.06GWh（Q3 增加 2.51GWh），其他规模较大的市场包括印度（前三季度累计 13.25GWh）、智利（前三季度累计 10.81GWh）。但新兴市场储能项目建设周期多为 2-3 年，装机落地速度较慢。整体来看，未来新兴市场将成全球储能增长重要引擎，带动全球储能需求持续攀升。

图表6：德国单月储能新增装机规模（MWh）



资料来源：ISRA 中信期货研究所

图表7：意大利单季度储能装机规模及数量



资料来源：ANIE 中信期货研究所

4. 三种情境下的储能需求预测

我们认为 2026 年中国与新兴市场将有较强的表现，欧美市场则会维持稳定的增长。

2026 年中国新增装机规模有望达到 230GWh。2026 年中国储能装机量有望保持和近期招标规模相近增速。结合 2025 年前三季度招标项目容量增速为 65%，在中国储能装机 140GWh 左右的基础上，2026 年年装机规模有望达到 230GWh，增长约 90GWh，贡献全球一半以上的装机量。

2026 年美国市场装机约为 52GWh。美国 EIA 预测 2026 年大储装机规模为 19.5GW/58.5GWh，假设美国储能完工率提升至 85%，则大储装机量接近 50GWh，工商业储能与户储预期保持与 2025 年相近需求（2GWh），总装机量为 52GWh，同比增长 11%。

预计 2026 年欧洲与新兴市场装机量约 167GWh。欧洲市场 2026 年的增长将主要来自各个国家大储市场的增长，德国、意大利有望分别提供 4GWh/3GWh 的增量，其他国家的增量则被计入新兴市场。新兴市场 2025 年底新招标规模约为 300GWh，考虑到施工进度较为缓慢，2026 年可兑现的装机量约为 40%，对应装机量 120GWh，叠加沙特 Tabuk 和 Hail 省储能项目（4GWh）、澳大利亚 AEMO 系统第三、四轮项目（4+3GWh）、维多利亚和南澳大利亚 600MW/2.4GWh 项目、智利 Colbún 项目（228MW/912MWh）等累积约 15GWh 前期项目，欧洲+新兴市场装机量大致为 167GWh，同比增长 52%。

预计 2026 年全球储能装机有望提升至 449GWh，储能电芯产量同步上调至 756GWh，这一最新预测远高于 2025 年初我们对市场的假想（此前预测分别为 369GWh 和 513GWh），对应碳酸锂需求增量 15 万吨，对于 2026 年碳酸锂供需格局或将产生重要影响。2025 年在国内外政策支持和市场需求拉动下，储能市场发展呈现超预期态势，我们上调对 2026 年储能装机量预测，主要来自两方面：一是中国三季度招采持续超预期，显示出容量电价激励下国内储能市场的蓬勃潜力；二是海外新兴市场（包括部分欧洲国家）需求进入高速增长期，大型项目招标不断。尽管美国、德国等传统市场表现较为平稳，但在两大新动能的驱使下，2026 年全球储能总需求增速将超过 50%，进而带动锂电相关材料进入高景气周期。

预计 2026 年储能项目的产能投放节奏较为均匀。按照历史规律，假设 2026 年需求高峰提前至 Q2，且高峰需求量接近 75-80GWh，仍有望使得电芯行业开工率达到 78% 以上。

同时，我们也关注到当下市场对于储能增速有所分歧，主要分歧在于快速增长的国内储能市场。为探讨 2026 年国内外储能需求的可能情形，我们基于不同的国内储能招标增速，做出两种假设推演（见图表 9）：

a)、激进情景：假设 2025H2-2026H1 国内储能招标增速达到 100%，这一增速水平接近当前新疆、内蒙等高收益率+高容量电价地区目前的招标增速，则 2026 年国内储能新增装机规模接近 280GWh，带动全球储能装机规模接近 500GWh，电芯产量接近 850GWh，对应碳酸锂需求 51 万吨；

b)、保守情景：假设 2026 年国内储能招标增速下降至 40%，这一增速为 Q3 全国储能招标增速水平，则 2026 年国内储能新增装机规模约为 200GWh，全球储能装机规模调整至 419GWh，电芯产量接近 700GWh，对应碳酸锂需求 42 万吨。

图表8：2021-2026 年全球储能新增装机预测

类别	单位	2021	2022	2023	2024	2025E	2026E		
							保守	中性	激进
中国市场储能装机	GWh	5	16	47	103	141	200	230	280
海外市场储能装机	GWh	25	50	70	104	157	219		
其中：美国	GWh	10	13	26	37	47	52		
欧洲+新兴	GWh	15	37	44	67	110	167		
全球储能装机合计	GWh	30	67	117	207	297	419	449	499
yoy	%		122%	75%	77%	43%	41%	52%	68%
全球储能电芯需求	GWh	66	145	193	347	499	700	756	850
yoy	%		119%	33%	80%	44%	40%	52%	70%
电芯备货系数		2.21	2.2	1.65	1.68	1.68	1.67	1.68	1.70
全球储能对应碳酸锂需求量	万吨	4	8.7	11.6	20.8	29.9	42.0	45.4	51.0

资料来源：中信期货研究所

储能电芯生产格局由中国主导，预计 2026 年中国储能电芯产量占比 96.5-97%，对应碳酸锂需求 43.8 万吨。储能电芯国产化率从 2021 年的 64%攀升至 2024 年的 96%-97%，标志着全球行业格局已从“中日韩三足鼎立”彻底演变为中国主导的格局。海外电池巨头（如 LG、三星 SDI）因坚守三元锂路线且成本高昂，已实质性退出储能电芯制造的主流竞争。而中国完全掌握了磷酸铁锂（LFP）产业链，从上游锂盐加工到正负极材料，拥有绝对的定价权和成本控制力；此外，中国企业率先推动 280Ah 向 314Ah、500Ah+ 升级，这种针对储能场景的专用化迭代速度远超海外，构建了技术代差。未来全球储能市场将长期维持“中国制造电芯+海外集成/组装”的分工格局。预期未来中国储能电芯产量占比将维持 96.5%-97% 的较高份额；中性预期下，2026 年中国储能电芯产量对应碳酸锂需求 43.8 万吨。

图表9：2021-2026 年储能电芯产量及对应碳酸锂需求

单位：GWh	单位	2021	2022	2023	2024	2025E	2026E		
							保守	中性	激进
全球储能电芯产量	GWh	66	145	193	347	499	700	756	850
其中：中国储能电芯产量	GWh	42	130	185	335	482	676	730	821
中国储能电芯产量占比	%	64%	90%	96%	97%	97%	97%		
全球储能电芯对应碳酸锂需求量	万吨	4	8.7	11.6	20.8	29.9	42.0	45.4	51.0
其中：中国	万吨	2.5	7.8	11.1	20.1	28.9	40.5	43.8	49.2

资料来源：中信期货研究所

我们对 2026 年储能电芯生产节奏进一步地按月度进行了拆分，以便全面分析 2026 年的碳酸锂的供需节奏变化，寻找年内供需错配机会。

图表10：2026 年分月度全球及中国储能电芯产量预测及对应碳酸锂需求测算

2026 年 (中性情景)	储能电芯产量(GWh)		对应碳酸锂需求量 (万吨)	
	全球	中国	全球	中国
1 月	51.0	49.5	3.06	2.97
2 月	53.0	51.4	3.18	3.09
3 月	55.0	53.4	3.30	3.20
4 月	58.0	56.3	3.48	3.38
5 月	61.0	59.2	3.66	3.55
6 月	64.0	62.1	3.84	3.73
7 月	67.0	65.0	4.02	3.90
8 月	70.0	67.9	4.20	4.08
9 月	70.0	67.9	4.20	4.08
10 月	69.0	66.9	4.14	4.02
11 月	69.0	66.9	4.14	4.02
12 月	69.0	66.9	4.14	4.02
2026 年合计	756	733.4	45.36	44.04

资料来源：中信期货研究所

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【Macro 宏观】

20251215 Insights from the Central Economic Work Conference 中央经济工作会议学习体会

20251210 Policy Readings on the Politburo Meeting in December 2025 年 12 月政治局会议联合点评

20251113 Commentary on China Monetary Policy Report Q3 2025 三季度货币政策执行报告点评

202501030 Policy Readings on the "15th Five-Year Plan" "十五五" 规划建议联合点评

20250801 Politburo Meeting Influence on Futures Prices 7 月政治局会议对期货价格影响

20250514 China-U.S. Tariff Suspension Influence on China Futures Prices 中美关税暂缓对中国期货价格影响

20250313 Influence of Geoenvironmental shift on commodities 地缘变局对商品价格影响分析

20250120 China Economic Annual Outlook 中国经济年度展望

20250114 Impact of "Two New" policy on China's Domestic Demand "两新" 政策对中国内需影响分析

20241109 Policy Readings of NPC Standing Committee Meeting in Nov. 11 月人大常委会新闻发布会解读

20241018 Policy Readings of China SCIO Press Conference on 17th Oct. 10 月 17 日国新办发布会解学习体会

20241013 Policy Readings of China Finance Ministry Press Conference on 11th Oct. 10 月财政部发布会学习体会

【Futures 期货】

20250723 Anti-Involution Policy Reading and Influence on Futures Prices "反内卷" 政策对期货价格影响

20250714 SHFE's Internationalization Reforms Interpretation 上海期货交易所国际化新规解读

20250622 China Accelerates Futures Market Opening Up 中国期货对外开放再提速

Research Framework 研究框架

【China Futures 中国期货】 Macro Economy/Assets Allocation /CTA/Equity Index/ Government Bond/Exchange Rate/ HK Equity index / Hang Seng Biotech Index/Crude Oil/Gold/Copper/Iron Ore/Steels/Lithium/Silicon Metal /LPG/PTA/LLDPE/Methanol/Urea/Shipping Freight /Live Hog /Soybean Meal /Soybean Oil/Palm Oil /Rapeseed Meal/Rapeseed Oil/Corn/Natural Rubber/Zinc/Silver

宏观经济/资产配置/CTA/股指/香港股指/国债/汇率/恒生生物科技指数/原油/黄金/铜/铁矿石/钢材/锂/工业硅/LPG/PTA/塑料/甲醇/尿素/航运/生猪/豆粕/豆油/棕榈油/菜粕/菜油/玉米/天然橡胶/锌/白银

【Global Futures 海外期货】 U.S. Economy / U.S. Treasury Bond/ U.S. Corn / US Dollar Index / BDI and FFAs

美国经济/美国国债/美国玉米/美元指数/航运指数

【Cross-Border Arbitrage 跨境套利】 Crude Oil/Iron Ore/Soybean/Copper/Rubber/Plam Oil 原油/铁矿石/大豆/铜/橡胶/棕榈油 Cross-Border Arbitrage Statistical Regression Strategy Model 跨境套利统计回归策略模型

【Risk Management 风险管理】 Iron Ore/Plastic/Steel/Freight/Fuel Oil 铁矿石/塑料/钢材/燃油/航运

【长期复盘】 全球大宗商品百年回顾/全球经济展望二十年大复盘/全球炼厂利润二十年回顾 /全球炼油产能中长期展望/世界石油海运要道/国际原油价格二十年复盘/美国天然气价格二十年复盘

【国别手册】亚太：中国/印度/日本/韩国/澳大利亚/新加坡/马来西亚/印度尼西亚；中东：沙特/伊朗/伊拉克/科威特/阿联酋/卡塔尔；美洲：巴西/委内瑞拉/圭亚那/加拿大；欧洲：俄罗斯/挪威/英国/法国；非洲：利比亚/尼日利亚/阿尔及利亚/安哥拉/刚果/南非/苏丹/南苏丹/埃及

Trading Manual 交易手册

Introduction on China Futures Market 中国期货市场概况

Introduction on China Futures Exchanges 中国期货交易所介绍

China Futures and Derivatives Law 中国期货法律法规

China Futures/Options Contracts Manual 中国期货合约手册

China QFII Policy Interpretation 合格境外投资者制度详解

China Futures Overseas Brokerage Policy Interpretation 中国期货境外经纪机构制度详解

Global Futures/Options Contracts Manual 海外期货/期权合约手册

【海外期货概况】股指/国债/汇率/能源/金属/农业

【境外交易所概况】中国香港/东南亚/欧洲/北美

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